higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS TECHNOLOGY L3
NQF LEVEL 3

12 NOVEMBER 2019

This marking guideline consists of 6 pages.

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SECTION A

QUESTION 1

1.1 B
1.2 C
1.3 D
1.4 A
1.5 C

(5 × 1) **[5]**

QUESTION 2

2.1 True
2.2 False
2.3 False
2.4 True
2.5 True

(5 × 1) **[5]**

QUESTION 3

3.1 Drying
3.2 Drum
3.3 Dehumidification
3.4 Adsorbent
3.5 Cation

(5 × 1) **[5]**

TOTAL SECTION A: 15

SECTION B

QUESTION 4

4.1 Chemical identity✓

- Indicates the name of the chemical, common names, alternative names and other methods of identifying a chemical✓

Physical and chemical characteristics✓

- Indicate properties of a chemical, such as bp, fp, density, vapour pressure, specific gravity, solubility, volatility, the product's general appearance and odour✓

Hazardous ingredients✓

- Indicates the hazardous properties of the chemical ✓

Firefighting measures✓

- Indicate chances that a fire or explosion can occur and the best ways of preventing them ✓

Reactivity data✓

- Shows other chemicals and substances with which it reacts and harzadous products that may form as a result of the reaction✓

Health hazards✓

- Show the health hazard signs and symptoms that may occur if a person has been exposed to the chemical✓

First aid measures✓

- Has information on what to do if a person comes into contact with a chemical and sometimes a doctor's note for treating the very same patient✓

Precautions for safe handling and use✓

- Precautions for safe handling, storage, use of chemical, methods for cleaning spills and leaks✓

Control measures✓

- Indicate engineering control, safe handling procedures and personal protective equipment (Any 5 × 2) (10)

4.2

- Mixing type
- Mixing time
- Material viscosity
- Turbulence
- Flow patterns (5)

4.3 4.3.1 Traditional shell and tube heat exchanger
4.3.2 Floating heat exchanger
4.3.3 Plate or plate and frame
4.3.4 Recuperator

(4 × 1) (4)

4.4
- Less uniform distribution over the bundle
- Has low and natural draught capability on fan failure
- Fans have higher noise level
- Have a lower heat removal compared to induced draught
- They are less controllable when compared to induced draught fans

(Any 4 × 1) (4)

4.5
- Circulating pumps
- Cooling towers
- Shell and tube heat exchangers (Any 2 × 1) (2)

[25]

QUESTION 5

5.1
- Counter-flow tower: the air enters the tower from the bottom and travels vertically up through the tower
- Cross-flow tower: the air enters the tower from various points on the side directly at right angles with the H_2O flow and then travels up through the tower. (2 × 2) (4)

5.2 5.2.1 Distributors: the distribution in a cooling tower distributes the hot water to the point where it has contact with air.✓ Cooling towers have different designs for water distribution.✓ Cross-flow towers use a high-pressure spray nozzle hot water distribution system to cover packing with water.✓ Hot water is distributed to the packing by gravity through metering orifices above the packaging.✓ Cross-flow towers generally require more energy for pumping the cross-flow towers✓ because there is no pressure spray distribution system in cross-flow systems.✓ (6)

5.2.2 Spray nozzles: cross-flow towers rely on nozzles to distribute water over the wet packaging deck.✓ Nozzles can either be fixed or sprayed in a round or square pattern✓ or can be part of the rotating assembly as found in some circular cross-flow towers.✓ There are different kinds of nozzles for each cooling tower type.✓ Many nozzles are made from PVC, ABS, polypropylene and glass-filled nylon.✓ (5)

5.3 CH_4 ✓ + $2O_2$ ✓ $\longrightarrow$ CO_2 ✓ + $2H_2O$✓✓

(1 mark for correct balancing) (5)

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5.4
- Food industries
- Chemical processing
- Petrochemial solids for devolatilisation
- Polymer and plastics
- Mineral and metal processes (5)

5.5 Because they do not change shape at high steam pressures and temperatures (2)

5.6
- Restricted or no flow of material through the dryer
- Irregular flow of the dry product through the dryer outlet
- Overdriven or burnt product at the end of the drying process
- Blocked filters containing excessive amounts of process material

(Any 3 × 1) (3)
[30]

QUESTION 6

6.1
- Silica gel
- Activated alumina
- Calcium oxide desiccant (3)

6.2
- Heatless desiccant dryer
- Heated
- Blower purge (3)

6.3
- Drying of gases
- Drying of solvents (2)

6.4 Solid material or ion exchanges used to remove contaminants from water (2)

6.5 Ion exchange has solid material known as ion exchange resins.✓ When water passes through them, ions from water pass through them.✓ The resin is in the form of granular beads.✓ It is usually packed into a bed, where water flows through the bed and parts the beads.✓ Ions from the water attach themselves to the beads, meaning that the water at the bottom of the bed is cleaner than water flowing into exchanges.✓ (5)

6.6 6.6.1 Synthetic organic ionic exchangers: they are the largest group of ion exchangers currently available.✓ They consist of a resin matrix with flexible network of hydrocarbon chains.✓ The network has fixed sites containing fixed ionic charges and the resins are made insoluble.✓ Synthetic organic ion exchangers have a high capacity for ion exchange.✓ Its advantage is that it has temperature limitations.✓ (5)

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6.6.2 Colloidal particles: Colloids are particles that range from 1nm to 1um.✓ Colloids are the most difficult to remove through mechanical separation means because of their size.✓ They can only be removed using a chemical.✓ Colloidal particles are pollutants found in water which must be removed.✓ Since the colloidal particles are unstable, they first need to be stabilised by the coagulation process in order to be removed✓ (5)

6.7
- Impurities
- Agitation
- Reagent dose
- Time
- Aeration conductivity
- High differential pressure
- Acidity and alkalinity of water (Any 5 × 1) (5)

[30]

TOTAL SECTION B: 85
GRAND TOTAL: 100

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